SKYFORCE — MATLAB Intercept Simulation: What It Should Show

Team 08, Anti-Drone Interceptor, Senior Track. This document describes the story the simulation should tell, not the code. It pulls together the problem statement and every design decision made so far, so whoever builds the MATLAB model knows what scenario to reproduce and what the plots need to prove.

1. Why this simulation exists

The problem statement asks for "analysis/simulation results" as part of the design report. A full Gazebo-style simulation is not realistic in a 2-day hackathon, and is not what is being asked for. What the judges need is proof that the numbers in the sizing calculations actually fly: that the interceptor can reach the target, that it has enough thrust and speed margin to do so, and that the mission fits inside the battery budget. A point-mass trajectory simulation in MATLAB does this well and is achievable in the time available.

The simulation exists to answer four questions:

1. Can the interceptor reach the intercept point before the target crosses the keep-out boundary?
2. Does it reach the target with enough closing speed and within the sizing model's thrust and speed limits?
3. Does the engagement fit inside the energy budget computed in the sizing document?
4. Does the post-capture decision (release on the spot, or tow to a safe point) work for a target captured outside the safety area?

2. Problem statement, restated for the simulation

The mission: detect, track and safely neutralise a rogue drone near an airport or event, with the target recovered intact. The requirements that the simulation must respect:

- Net capture, with parachute-assisted recovery of targets up to 5 kg.
- Sprint speed of at least 25 m/s.
- Detection range of at least 1 km, via the stated sensor (ground radar).
- Vision or radar terminal guidance.
- Takeoff weight under 25 kg.

The simulation should demonstrate the capture mechanism, the guidance approach, and how collateral risk is kept minimal, which are the three things the problem statement explicitly asks the team to justify.

3. The scenario the simulation should recreate

This is the mission timeline agreed on so far, and the simulation should step through it in order:

Detection and cueing. The ground radar detects the target at some distance (at least 1 km) and the ground control station computes a target position and velocity. The interceptor is cued to a predicted intercept point. The simulation does not need to model the radar itself, but it should start from a radar-style initial condition: a target position, heading and speed at a given range, with a reaction delay before the interceptor launches.

This reaction delay should be built up from the components already named in the architecture document, rather than picked arbitrarily: the radar's own track update rate (the TrueView R30 reference is configurable from 64 ms to 3 s; a mid value such as 0.5 s is a reasonable assumption), plus GCS processing and intercept-point planning (assumed, on the order of 0.3 to 0.5 s), plus the long-range radio link latency carrying the cue up to the drone (assumed, under 0.1 to 0.2 s for the link itself), plus the interceptor's own spin-up and launch decision time (assumed, 0.5 to 1 s for motors to reach flight RPM and the flight controller to accept the guided-mode command). Summed, this gives a working assumption of roughly 1.5 to 2 seconds of total reaction delay before the interceptor is airborne and closing. This is a stacked assumption, not a measured figure, and should be replaced with real component timings once a specific radar and radio are chosen.

Mode selection. Based on target speed relative to the interceptor's own 25 m/s capability, the simulation should pick one of two engagement geometries:

- Attack mode: the target is slower than the interceptor, so the interceptor chases and closes in from behind or from an angle.
- Defense mode: the target is faster than the interceptor, so the interceptor cannot out-run it. Instead it moves to a lead point ahead of the target's predicted track and waits for the target to fly into range.

The simulation should include at least one case of each mode, since this was a specific concern raised during design: what happens if the target is faster than the interceptor.

Guidance. Once within engagement range, the interceptor closes the final distance using a standard guidance law (proportional navigation is the natural choice, since it is what vision and radar terminal guidance normally use). The simulation should show the interceptor's path curving onto a collision course with the target, not flying straight at the target's current position.

Net engagement window. The problem statement and the design decisions so far set a design-target firing range of 5 to 15 m. The simulation should mark the point where the interceptor enters this window with a stable closing geometry, since this is the moment capture is assumed to happen. The simulation does not need to model the net physics itself (the mesh, the propellant, the fouling of the target's rotors) — that is outside what a point-mass trajectory model can show. It only needs to show that the interceptor can arrive in that window with a controllable approach.

Post-capture decision. After the simulated capture point, the combined mass of interceptor and target (up to 23.5 kg, from the sizing document) should be used for a second, shorter phase: the interceptor checks its position against a pre-set safety boundary.

- If capture happens inside the safe area, the simulation ends with an immediate landing/release at that point.
- If capture happens outside the safe area, the simulation should show the interceptor, now carrying the extra mass, flying to the nearest pre-defined safe point before release. This is the tow phase already sized in the calculations document (about 60 seconds, at the heavier all-up mass).
  The simulation should run this as two scenarios, not one: one where the target is captured inside a designated safety point (immediate release, no tow), and one where it is captured near a point that is not a safety point (tow required). This tests both branches of the release-or-tow decision rather than assuming only one of them happens.

Abort case. The simulation should also include one run where the engagement is aborted. The abort decision should check two conditions together, not just one:

- Hard boundary: has the target's position crossed the keep-out boundary around the airport or event?
- Time-to-impact: given the current closing geometry, will the time remaining before the target reaches the boundary run out before the interceptor can reach the predicted intercept point?

The engagement aborts if either condition is triggered. This demonstrates the "safe abort" rule from the design discussion, where the team deliberately chose not to promise a 100% capture rate.

4. What physical limits the simulation should respect

The simulation should use the limits already established in the sizing calculations, so the two documents agree with each other:

- Maximum interceptor speed: 25 m/s.
- Maximum thrust-to-weight ratio: 2.43 (six motors, 7.3 kgf max thrust each).
- Sprint power and the implied acceleration limit, so the interceptor cannot instantaneously jump to 25 m/s — it should show a realistic acceleration profile.
- Mission energy budget: about 207 Wh for an intercept-only engagement, or about 253 Wh if a tow phase is needed. The simulation should track elapsed flight time and flag whether a given run would exceed the available battery energy.
- Engagement radius: Fortem's reference mission covers roughly 0.5 to 1.5 km; the team's working assumption is 1.5 to 2 km.

5. What the simulation should output

For each scenario run (attack mode, defense mode, tow-required, abort), the simulation should produce:

- A 2D (or simple 3D) trajectory plot showing both the interceptor's and the target's paths over time, with the capture point marked.
- A closing-distance-versus-time plot, showing the range between interceptor and target shrinking to the net engagement window.
- A speed-versus-time plot for the interceptor, to check it stays within the 25 m/s and acceleration limits used in the sizing document.
- A simple energy-versus-time tally, compared against the 207 Wh / 253 Wh budget, so a reviewer can see at a glance whether the mission fits inside the battery sized earlier.
- A short summary table per scenario: initial range, target speed, mode selected, time to intercept, miss distance at closest approach, and pass/fail against the energy budget.

6. What the simulation is not expected to show

To keep expectations honest for the report, the simulation should not claim to demonstrate:

- The net's physical deployment, opening, or how reliably it fouls the target's rotors.
- The parachute's opening shock, canopy dynamics, or wind drift.
- Sensor noise, false tracks, or radar detection probability.
- Structural loads, vibration, or motor thermal behaviour.
- Collision avoidance with other aircraft or obstacles.

These are the same limitations already noted in the sizing and architecture documents. The simulation proves the guidance and energy budget close; it does not prove the capture and recovery hardware works, which is a separate (and currently unvalidated) part of the design.

7. Open questions for the team

- What proportional navigation gain (or alternative guidance law) should be used? Resolved: N = 3, a common default for this kind of intercept guidance. The simulation should still check that this gain's turns stay inside the sizing model's thrust margin, and adjust it if not.
- Reaction delay between detection and launch: now assumed as roughly 1.5 to 2 seconds, built up from the radar update rate, GCS planning time, radio latency and interceptor spin-up (Section 3). To be replaced with real figures once components are chosen.
- Safe points for the tow-phase scenario: resolved. Two scenarios are run — one where capture happens inside a designated safety point (immediate release), and one where it happens near a non-safety point (tow to the nearest safe point required).
- Abort rule: resolved. The abort check combines both a hard keep-out boundary and a time-to-impact threshold; the engagement aborts if either is triggered (Section 3).